Audio Forensic Intelligence Platform

Project Documentation — Version 1.0

Version	1.0
Status	Planning / Development
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Prepared by	Intern — Forensic Systems Project

1. Project Overview

The Audio Forensic Intelligence Platform is a web-based tool for verifying the authenticity of audio recordings. It helps analysts determine whether a recording has been tampered with, identify where modifications occurred, and generate forensic reports for investigation and evidence review.

The system combines audio fingerprinting, digital watermarking, tamper detection, and PDF report generation in one interface.

Problem Being Solved

Audio recordings used as evidence can be altered with widely available editing tools. This platform addresses three core needs:

- Detecting whether an audio file has been modified
- Identifying which segments were altered and with what confidence
- Producing a standardized report for investigation and evidence review

Project Objectives

Objective	Purpose
Audio Fingerprinting	Generate a unique identity signature for each recording
Digital Watermarking	Embed hidden ownership data; verify it later
Verification	Compare suspect audio against the registered original
Tamper Detection	Detect signs of insertion, deletion, or replacement
Tamper Localization	Identify exact timestamps where tampering occurred
Similarity Analysis	Measure percentage similarity between two recordings
PDF Report Generation	Produce a downloadable forensic investigation report
User Access Control	Role-based access for analysts, investigators, and admins

2. Project Scope — Version 1

The table below separates what will be built during the internship from what is planned for future versions.

Phase 1 — Internship Deliverables	Future Enhancements
Audio Upload (WAV / MP3)	AI Deepfake Detection
Audio Fingerprinting	Blockchain Evidence Tracking
Digital Watermarking	Real-Time Verification
Verification Module	Cloud Storage Integration
Tamper Detection	Multi-user Investigation Workflows
Tamper Localization	Advanced Forensic Analytics
Similarity Analysis	
PDF Report Generation	

3. User Types & Permissions

Role	Responsibilities	Permissions
Forensic Analyst	Uploads audio, runs analysis, generates reports	Upload files, run all modules, download PDF reports
Investigator	Reviews findings and evidence records	View reports and analysis results — read only
System Administrator	Manages users, cases, and system settings	User management, case management, system monitoring

4. System Architecture

The platform is organized in five layers. Each layer communicates only with the one directly below it.

Layer	Technology	Role
User / Browser	HTML, CSS, JavaScript, Bootstrap	Interface and user interaction
Django Web Application	Python, Django (ASGI), DRF	Routing, authentication, API endpoints
Audio Processing Engine	Librosa, NumPy, SciPy	Fingerprinting, watermarking, tamper detection
Database	SQLite (dev) / PostgreSQL (prod)	Persistent storage for all records and reports
PDF Report Generator	ReportLab, Matplotlib	Compiles findings into downloadable PDF

5. Database Entities

Core tables the system needs to function.

Entity	Purpose
User	Login credentials and assigned role
AudioFile	Uploaded file details — name, format, duration, case ID
Fingerprint	Generated fingerprint hash for each audio file
Watermark	Embedded watermark data — case ID, owner, timestamp
VerificationRecord	Analysis results — similarity score, tamper verdict, segments
Report	Generated PDF — linked to verification record

6. Core Features

Module	What it does	Output
6.1 Audio Upload	Accepts WAV and MP3 files; records metadata and assigns a Case ID	File stored with metadata
6.2 Fingerprinting	Generates a unique cryptographic signature for the audio signal	Fingerprint hash saved to DB
6.3 Watermarking	Embeds hidden ownership data (Case ID, analyst, timestamp) into the audio	Watermarked audio file
6.4 Verification	Compares suspect file fingerprint against the registered original	Authentic / Potentially Modified
6.5 Tamper Detection	Scans for signal anomalies — insertions, deletions, replacements	Tampered / Not Tampered
6.6 Tamper Localization	Identifies exact timestamp ranges where tampering occurred	e.g. 00:13–00:18, 92% confidence
6.7 Similarity Analysis	Calculates a percentage match between two recordings	e.g. 94.5% similarity
6.8 Report Generation	Compiles all findings into a downloadable PDF	Forensic PDF report

7. System Workflow

7.1 Registering an Original Recording

Step	Action
1	Analyst uploads the original audio file and assigns a Case ID
2	System generates fingerprint and embeds watermark automatically
3	All data saved to the database

7.2 Analyzing a Suspect Recording

Step	Action
1	Analyst uploads the suspect file and links it to the existing Case ID
2	System extracts fingerprint and verifies watermark integrity
3	Fingerprint compared against original; tamper detection runs
4	Modified segments identified with timestamps and confidence scores
5	Full PDF forensic report generated and made available for download

8. How to Use the Platform

Forensic Analyst

- Log in and navigate to Audio Upload
- Upload the original file, assign a Case ID, and select Register Original
- To analyze a suspect file: upload it, link to the Case ID, select Verify & Analyze
- Review results on the dashboard and click Generate Report to download the PDF

Investigator

- Log in and go to Reports or Case Management
- Select a case to view analysis results, segment findings, and similarity scores
- Download PDF reports as needed

Administrator

- Manage user accounts and assign roles from the Admin panel
- Create and archive investigation cases
- Monitor system activity logs and storage usage

9. Expected Challenges

Challenge	Mitigation Plan
Accurate tamper localization	Use sliding-window analysis; tune confidence thresholds during testing
MP3 compression artifacts	Apply format-aware preprocessing before analysis
Watermark robustness	Use frequency-domain embedding that survives common re-encoding
Large file processing	Process audio in chunks; use async tasks if needed
Reducing false positives	Calibrate detection thresholds using a labeled test dataset
Delivering all 8 modules on time	Prioritize fingerprinting, verification, and reporting as the baseline

10. Success Criteria

The project is considered successfully delivered when all of the following are working:

Module	Success Condition
Audio Upload	WAV and MP3 files upload correctly; metadata stored in database
Fingerprinting	Unique fingerprint generated for every uploaded file
Watermarking	Watermark embedded and extractable for verification
Tamper Detection	Modified recordings correctly identified
Tamper Localization	Suspicious segments reported with timestamps and confidence scores
Similarity Analysis	Percentage similarity score produced for any two recordings
Report Generation	Complete PDF report downloadable after every analysis run
Access Control	Each role can only access its permitted features